Building Strategic Resilience: Lessons from an Industrial Automation Company

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Abstract

Strategic resilience has become an essential capability for organisations operating in increasingly uncertain environments. This paper examines how an industrial technology company can strengthen its resilience while preserving the competitive advantages that underpin its long-term success. Using the case of Company A, the study combines external and internal strategic analysis to assess the alignment between the company's operating model and its evolving business environment. The analysis shows that while Company A maintained a strong competitive position throughout the COVID-19 pandemic, the crisis exposed structural dependencies related to international mobility and on-site service delivery. To address these vulnerabilities, the paper proposes three complementary strategic priorities: market diversification, increased customer autonomy, and the development of data-driven services. The findings suggest that long-term resilience depends not only on sustaining competitive advantage, but also on continuously adapting the mechanisms through which value is created and delivered.

Introduction

Operational excellence has long been a source of competitive advantage for industrial companies. Continuous improvement, automation, and globalization have enabled organizations to increase productivity, reduce costs, and expand into international markets. However, they have also increased dependence on complex supply chains, global mobility, and interconnected operations.

The COVID-19 pandemic demonstrated how quickly these dependencies could become strategic vulnerabilities. Companies with strong financial performance and efficient operations still faced significant disruption when travel restrictions, supply chain interruptions, and market uncertainty altered the conditions on which their business models relied. Indeed, the crisis renewed attention to strategic resilience, defined as an organization's capacity to sustain value creation despite major external shocks.

This paper examines strategic resilience through the case of Company A, a French manufacturer of high-precision automated systems for the fragrance and flavour industries. The company entered the pandemic in a strong competitive position. Yet, the crisis exposed several structural vulnerabilities within its business model. The analysis identifies these dependencies and discusses practical actions that can strengthen resilience before the next macroeconomic shock.

Company Background

Company Overview

Company A is a French industrial technology company founded in 1996. The company designs, manufactures, and services high-precision automated dispensing systems used primarily by manufacturers in the fragrance and flavour industries. Its solutions automate weighing, dosing, storage, and production processes, enabling customers to improve accuracy, productivity, and product consistency while reducing manual operations. In addition to its equipment, the company develops software to control and monitor its systems and provides engineering, installation, maintenance, and long-term technical support.

Unlike manufacturers offering standardised equipment, Company A has built its business around highly customised solutions tailored to each customer's production environment. Its machines are designed using a modular architecture based on standard building blocks that can be combined to create more complex systems. This approach simplifies manufacturing while providing the flexibility required to address a wide variety of customer needs. It also creates a continuous feedback loop between engineering and production, allowing new customer requirements to be incorporated into future designs.

At the time of the study, Company A employed 24 people and generated annual revenues of approximately €8 million. Although relatively small, the company had established itself as a recognised international player within its niche, with around 180 installations worldwide and almost one quarter of its revenue generated through exports. Its customer base consisted primarily of manufacturers seeking to automate production while maintaining extremely high levels of precision and reproducibility.

A distinctive feature of Company A's business model is its emphasis on long-term customer relationships. Equipment sales represent the beginning rather than the end of the commercial relationship. Annual service contracts covering both hardware and software support provide recurring revenue while creating strong incentives to design reliable, durable machines that remain operational over many years. This close collaboration also enables the company to gain a detailed understanding of customer operations, creating opportunities for continuous product improvement and future service innovation.

Company Performance

Before the COVID-19 pandemic, Company A was in a strong competitive position. Between 2016 and 2018, revenue grew at an annualised rate of 24%, while operating income and net income increased by 44% and 40% respectively. Profitability ratios also reflected a highly efficient business model, with a net margin of 27%, a return on assets of 20%, and a return on equity of 26%. These results indicate that profitability improved faster than revenue, suggesting that the company successfully scaled its operations while maintaining tight control over costs.

The company's financial position was equally robust. Liquidity ratios remained well above conventional benchmarks, with a quick ratio of 3.24 and a current ratio of 3.60. Interest coverage exceeded 150, reflecting a very limited reliance on debt financing. Rather than using leverage to accelerate growth, Company A had financed its expansion primarily through internally generated cash flows, giving it substantial financial flexibility entering the pandemic.

Overall, Company A entered the COVID-19 crisis from a position of strength. It operated in an attractive niche, possessed distinctive technological capabilities, maintained long-term customer relationships, and demonstrated consistent financial performance. The strategic challenge examined in this paper therefore does not stem from weak fundamentals. Instead, it arises from the way the company's operating model interacts with an external environment that changed abruptly in 2020. This distinction is important because it shifts the focus of the analysis from business recovery to strategic resilience.

Despite its strong competitive position, the COVID-19 pandemic exposed important vulnerabilities within the company's operating model, raising the central question addressed in this paper.

Situational Analysis

The strategic challenge presented in the previous section cannot be understood through the pandemic alone. A comprehensive assessment of the external environment, industry dynamics, and internal capabilities is required to distinguish temporary disruptions from structural issues affecting the company's long-term competitiveness. This situational analysis therefore combines several complementary strategic frameworks to evaluate both the opportunities available to Company A and the constraints shaping its future development. The findings provide the foundation for the strategic priorities and recommendations presented in the following chapters.

External Environment

Macro-Environmental Analysis

The external environment was assessed using the ESTEMPLE framework to identify the political, economic, social, technological, environmental, legal, and ethical factors most likely to influence Company A's future development. Although the framework considers a broad range of external forces, only a limited number were expected to have a significant strategic impact.

Factor	Opportunity / Threat	Strategic Implication
COVID-19 travel restrictions	Threat	Reduced ability to install and service equipment internationally
Growing demand for industrial automation	Opportunity	Sustained market growth for automated dispensing solutions
Industry 4.0 technologies	Opportunity	Expansion of connected equipment and digital services
Resilience of fragrance and flavour markets	Opportunity	Stable long-term demand despite the pandemic

Travel restrictions represented the most immediate threat because they directly affected installation, commissioning, and technical support activities. However, the broader outlook remained favourable. Demand for industrial automation continued to increase as manufacturers sought greater productivity, higher process reliability, and reduced dependence on manual operations. At the same time, Industry 4.0 technologies created new opportunities to complement physical equipment with connected services, remote monitoring, and predictive maintenance. Finally, the company's primary customer segments remained relatively resilient throughout the pandemic, limiting the impact on underlying market demand.

Although the macro-environment remained favourable, these observations provide only a partial understanding of Company A's strategic position. The attractiveness of the industry must also be assessed.

Industry Analysis

The attractiveness of the industry was subsequently assessed using Porter's Five Forces framework. The analysis indicates that Company A operates within a specialised market characterised by relatively high barriers to entry and limited direct competition.

Technological expertise, customer-specific engineering, and long development cycles restrict the arrival of new competitors. Existing rivalry also remains moderate, as the market consists of a limited number of established international suppliers competing primarily through technological performance and customisation rather than price. Customer bargaining power is constrained by the high level of integration between dispensing systems and production processes, making supplier substitution costly once equipment has been installed. Supplier bargaining power is similarly limited because most components can be sourced from multiple providers. Finally, although manual dispensing represents a potential substitute, it cannot deliver the precision, repeatability, or productivity required for industrial-scale manufacturing.

An attractive industry alone does not guarantee sustained competitive advantage. The next section therefore examines the internal capabilities that differentiate Company A from its competitors.

Internal Analysis

While the external analysis confirms that Company A operates in an attractive market, long-term resilience ultimately depends on the firm's own resources and capabilities. This section examines how value is created within the organization, identifies the sources of its competitive advantage, and assesses its ability to adapt to changing market conditions.

Value Chain Analysis

Company A creates value through an integrated business model combining engineering, manufacturing, software development, installation, and long-term technical support. Unlike manufacturers of standard industrial equipment, the company delivers highly customised solutions designed around each customer's production environment. Value is therefore generated throughout the entire customer lifecycle rather than at the point of sale alone.

Primary Activity	Strategic Contribution
Research & Development	Continuous innovation and product differentiation

Primary Activity	Strategic Contribution
Engineering & System Design	Customised solutions adapted to customer requirements
Manufacturing	High product quality and precision
Installation & Commissioning	Integration into customer production processes
After-Sales Service	Customer retention and recurring service revenue

Several observations emerge from this analysis. First, the company's competitive advantage is built well before production begins. Engineering expertise, product design, and customer collaboration represent the most significant sources of value creation and are difficult for competitors to replicate. Second, the modular architecture of the company's products enables a high degree of customisation while limiting manufacturing complexity. This balance between flexibility and standardisation contributes both to operational efficiency and customer satisfaction.

The analysis also reveals an important structural dependency. Although most activities can be performed from the company's facilities, installation, commissioning, and certain maintenance operations require engineers to travel to customer sites. These activities therefore became vulnerable when international travel was restricted during the pandemic. The crisis did not weaken the company's capabilities, but it temporarily constrained part of the mechanism through which those capabilities created value.

Competitive Advantage

Beyond the activities that create value, Company A's competitive position is supported by a combination of strategic resources and organisational capabilities that are difficult for competitors to replicate.

One of the company's principal strengths lies in its engineering expertise. Over many years, it has developed specialised knowledge in designing automated dispensing systems tailored to the complex requirements of the fragrance and flavour industries. This expertise enables the company to deliver highly customised solutions while maintaining high standards of precision, reliability, and quality.

A second source of competitive advantage is its modular product architecture. By designing equipment around reusable technological building blocks, Company A can adapt its solutions to diverse customer requirements without redesigning each system from the ground up. This approach shortens development times, improves manufacturing efficiency, and facilitates future upgrades, providing both operational and commercial flexibility.

The company also benefits from long-standing relationships with customers built through the installation, maintenance, and continuous improvement of its equipment. These relationships create a deep understanding of customer processes, strengthen trust, and generate opportunities for repeat business and long-term service contracts. Such customer proximity represents an important intangible asset that is difficult for new entrants to reproduce.

Finally, the company's strong financial performance provides the resources necessary to support innovation and long-term strategic investments. Stable profitability and a solid financial position increase the organisation's capacity to adapt its products, expand into new markets, and respond to changing customer expectations without compromising operational stability.

Taken together, these resources form a coherent system of competitive advantage. Applying the VRIO framework suggests that they are valuable, supported by the organisation, and sufficiently difficult to imitate due to the accumulation of technical expertise, customer knowledge, and industry experience over time. Rather than relying on a single distinctive capability, Company A's competitive position results from the interaction of complementary resources that reinforce one another and provide a strong foundation for future strategic development.

Dynamic Capabilities

Beyond its existing strengths, the company's long-term resilience depends on its ability to adapt as its environment evolves.

Several characteristics suggest that Company A possesses strong dynamic capabilities. Its modular engineering approach facilitates continuous product evolution, while close collaboration between technical teams and customers enables rapid identification of emerging needs. The company's history of incremental innovation also demonstrates an ability to integrate new technologies without disrupting existing operations.

However, the pandemic highlighted an opportunity to extend these adaptive capabilities beyond product development. While the company had successfully evolved its products over time, less attention had been given to evolving the way those products were delivered and supported. Strengthening remote service capabilities, increasing customer autonomy, and developing digital services therefore represent logical extensions of capabilities that already exist within the organisation.

Key Findings

The strategic analysis highlights three principal findings that shape the remainder of this paper.

First, Company A operates in a favourable external environment. The continued expansion of industrial automation, the growing adoption of Industry 4.0 technologies, and the resilience of the fragrance and flavour industries provide strong long-term growth opportunities. The competitive environment also remains attractive, characterised by high barriers to entry and limited competitive rivalry, allowing specialised companies with differentiated capabilities to sustain their market position.

Second, the company possesses significant internal strengths that support its competitive advantage. Its engineering expertise, modular product architecture, long-standing customer relationships, and strong financial performance provide a solid foundation for future growth. These capabilities have enabled the company to develop highly customised solutions while maintaining a reputation for quality and reliability in a specialised industrial market.

Finally, the analysis reveals that the company's primary vulnerability does not lie in its strategy or market positioning, but in the way value is delivered to customers. The COVID-19 pandemic exposed the organisation's dependence on international travel and on-site technical interventions for installation, commissioning, and maintenance activities. While these dependencies had little impact under normal operating conditions, they reduced operational flexibility during periods of severe disruption.

Taken together, these findings suggest that Company A does not require a fundamental strategic repositioning. Instead, its long-term resilience depends on strengthening the alignment between its existing competitive advantages and an evolving business environment. The following chapter examines this question through the concept of strategic fit, identifying the strategic priorities that will enable the company to build upon its existing strengths while reducing its exposure to future disruptions.

Strategic Fit

The situational analysis demonstrates that Company A's competitive position remains fundamentally strong. The company operates in an attractive industry, possesses valuable technical capabilities, and continues to benefit from a differentiated market position. However, the pandemic exposed several structural dependencies that could limit its ability to respond to future disruptions. The challenge is therefore not to redefine the company's strategy, but to ensure that it remains aligned with a changing business environment.

This section integrates the findings from the previous analysis to identify the strategic priorities that should guide the company's future development. It also assesses whether the current strategy is sufficient to maintain long-term competitiveness or whether strategic adjustments are required to avoid strategic drift.

Strategic Imperatives

The external and internal analyses converge towards three strategic imperatives that are consistent with both the company's existing capabilities and the opportunities emerging within its environment.

The first imperative is market diversification. Company A has established a strong position within the fragrance and flavour industries, but this concentration increases its exposure to sector-specific fluctuations. At the same time, several adjacent industries require similar levels of precision, automation, and process control. The company's engineering expertise and modular product architecture provide a strong foundation for expanding into these markets with relatively limited adaptation. Diversification therefore represents an opportunity to strengthen resilience while leveraging existing capabilities rather than developing entirely new ones.

The second imperative is increasing customer autonomy. The value chain analysis highlighted that installation, commissioning, and technical support remain highly dependent on on-site interventions. Although this approach has historically contributed to customer satisfaction, the pandemic demonstrated the risks associated with relying extensively on international travel. Strengthening remote support capabilities, improving technical documentation, expanding customer training, and simplifying maintenance procedures would reduce this dependency while improving operational flexibility for both the company and its customers.

The third imperative is leveraging operational data. The analysis of the external environment showed that Industry 4.0 technologies continue to reshape industrial manufacturing. Connected equipment, remote monitoring, and predictive maintenance create opportunities to extend customer relationships beyond equipment delivery. Company A already possesses the engineering expertise and installed customer base required to support this evolution. The challenge lies in transforming operational data into services that generate additional value while complementing the company's existing business model.

These three imperatives are closely interconnected. Diversification expands growth opportunities, customer autonomy reduces operational constraints, and digital services strengthen long-term customer relationships. Together, they address the principal vulnerabilities identified during the situational analysis while reinforcing the capabilities that have historically differentiated Company A.

Assessing the Risk of Strategic Drift

Johnson, Scholes, and Whittington describe strategic drift as the gradual misalignment between an organization's strategy and its external environment. Rather than resulting from a sudden decline in performance, strategic drift often emerges when organizations continue to rely on strategies that have historically been successful despite significant changes in their competitive environment.

At first glance, Company A shows few of the characteristics commonly associated with strategic drift. The company remains profitable, financially sound, and well positioned within an attractive market. Its products continue to meet customer needs, demand for industrial automation remains strong, and its engineering capabilities continue to differentiate it from competitors. None of the analyses conducted in this study suggest that the company's competitive position is deteriorating.

However, the pandemic revealed early signals that the operating model may become increasingly exposed if it evolves more slowly than its environment. The value chain remains highly dependent on physical interventions, while advances in digital technologies are creating new expectations regarding remote support, predictive maintenance, and connected services. At the same time, continued concentration within a limited number of customer segments increases exposure to industry-specific disruptions.

These observations do not indicate that Company A is experiencing strategic drift today. Instead, they suggest that the conditions under which its strategy has historically succeeded are evolving. If these changes continue without a corresponding adaptation of the operating model, the gap between market expectations and the company's methods of creating and delivering value is likely to widen over time.

The objective is therefore not to replace the current strategy, but to extend it. The company's existing competitive advantages remain highly relevant and provide a strong platform for future growth. Strengthening resilience requires preserving these advantages while adapting the business model to reduce structural dependencies and exploit new opportunities created by technological and market developments.

This assessment supports an evolutionary approach to strategy. Rather than pursuing a radical transformation, Company A should build upon its existing strengths by broadening its market presence, increasing customer autonomy, and progressively expanding its digital capabilities. These strategic directions preserve the company's competitive identity while improving its ability to adapt to future disruptions.

Strategic Recommendations

The previous analysis identified three strategic imperatives for strengthening Company A's long-term resilience: diversifying its sources of growth, reducing operational dependencies, and expanding the value created for existing customers. These recommendations build upon the company's existing capabilities and are designed to reinforce its competitive position without fundamentally altering its business model.

Rather than pursuing a single transformational initiative, the proposed strategy combines several complementary actions that can be implemented progressively. Together, they reduce the company's exposure to future disruptions while creating new opportunities for sustainable growth.

Strategic Posture

Before defining specific initiatives, it is important to determine the overall strategic posture that should guide their implementation.

Company A operates in an attractive industry and continues to possess strong competitive advantages. Consequently, the objective is not to replace the existing strategy, but to extend it. The company should preserve the capabilities that have historically driven its success while adapting its operating model to better reflect changes in its external environment.

This approach is consistent with an **adaptive strategy**. Rather than reacting to crises through isolated corrective actions, the company should progressively strengthen its resilience by investing in initiatives that remain valuable regardless of future disruptions. Such "no-regret" decisions improve operational flexibility, broaden growth opportunities, and reinforce customer relationships even if similar crises never occur again.

Market Diversification

The first recommendation is to diversify the company's sources of revenue by expanding into carefully selected industries and geographic markets.

Diversification reduces dependence on a limited customer base while allowing the company to leverage its existing engineering expertise and modular product architecture across a wider range of industrial applications. Since the company's core capabilities remain transferable, expansion into adjacent markets requires significantly lower investment than developing entirely new technologies.

Several industries were evaluated using a weighted decision matrix incorporating criteria such as market attractiveness, technical compatibility, implementation complexity, competitive intensity, profitability, and strategic fit.

[**Insert Industry Decision Matrix**]

The analysis identified **XXX** as the most attractive industry for expansion, combining strong growth potential with a high degree of alignment with Company A's existing capabilities.

A similar evaluation was conducted to identify priority geographic markets.

[**Insert Country Decision Matrix**]

The selected countries provide attractive growth opportunities while remaining consistent with the company's available resources and international development capabilities.

Increasing Customer Autonomy

The second recommendation focuses on reducing the company's dependence on physical interventions by enabling customers to perform a greater proportion of routine activities independently.

Several complementary initiatives are proposed, including enhanced technical documentation, expanded customer training, modular installation procedures, remote diagnostics, and improved monitoring capabilities. Together, these measures reduce reliance on international travel while improving responsiveness and service continuity.

Increasing customer autonomy also creates operational benefits beyond resilience. Faster issue resolution, lower service costs, shorter response times, and improved customer satisfaction all contribute to strengthening the company's competitive position while reducing pressure on internal engineering teams.

Expanding Data-Driven Services

The third recommendation aims to capture additional value from the company's installed base by progressively expanding digital services.

The continued adoption of connected industrial equipment creates opportunities to complement hardware sales with remote monitoring, predictive maintenance, performance analytics, and decision-support tools. These services strengthen customer relationships while generating recurring revenue streams that are less dependent on new equipment sales.

Importantly, this recommendation does not require a fundamental change in the company's identity. Instead, it extends the existing value proposition by combining engineering expertise with digital capabilities that are becoming increasingly important within industrial manufacturing.

Implementation Roadmap

The proposed recommendations should be implemented progressively in order to balance investment requirements, organisational capacity, and expected benefits.

In the short term, priority should be given to market diversification and initiatives that increase customer autonomy, as these directly address the principal operational dependencies revealed during the pandemic.

Once these foundations have been established, the company should progressively expand its digital service offering by leveraging the operational data generated through its installed equipment. This phased approach allows Company A to strengthen resilience while maintaining strategic focus and operational stability.

Discussion

The case of Company A demonstrates how major external disruptions can reveal strategic vulnerabilities that remain largely invisible during periods of stability. Although the COVID-19 pandemic served as the catalyst for this analysis, the findings extend beyond a single crisis and offer broader insights into how industrial technology companies can strengthen their long-term resilience.

Managerial Implications

One of the principal insights from this study is that strategic resilience should not be viewed solely as a matter of risk management. Traditional resilience approaches often focus on preparing organisations to respond to unexpected events through contingency planning, redundancy, or crisis management. While these mechanisms remain important, the present case suggests that resilience is equally determined by the design of the business model itself.

Company A entered the pandemic with strong financial performance, recognised technical expertise, and a favourable competitive position. None of these competitive advantages were fundamentally weakened by the crisis. Instead, the disruption exposed assumptions embedded within the company's operating model, particularly its reliance on international travel and physical customer interactions to deliver value.

This distinction has important implications for managers. Building resilience does not necessarily require replacing an existing strategy or pursuing radical organisational transformation. In many cases, resilience can be strengthened by systematically identifying operational dependencies and progressively reducing them while preserving the capabilities that continue to differentiate the organisation.

The recommendations developed in this paper reflect this perspective. Market diversification reduces dependence on a limited customer base, greater customer autonomy improves operational flexibility, and data-driven services extend the company's value proposition while creating additional recurring revenue. Although these initiatives address vulnerabilities revealed during the pandemic, they also strengthen the company's competitive position under normal operating conditions. Consequently, they represent strategic investments that create value regardless of whether similar disruptions occur in the future.

More broadly, the case highlights the importance of regularly reassessing not only where competitive advantage originates, but also how that advantage is delivered. As customer expectations, digital technologies, and global operating conditions continue to evolve, maintaining this alignment is likely to become an increasingly important source of sustained competitive advantage.

Limitations

This study is based on a single case study and therefore reflects the characteristics of one industrial technology company operating within a specific industry and historical context. Although the analytical framework adopted in this paper can be applied more broadly, the strategic recommendations should not be considered universally applicable without taking into account the specific circumstances of other organisations.

In addition, the analysis was conducted in the context of the COVID-19 pandemic, when significant uncertainty surrounded both the duration of the disruption and its long-term economic consequences. Different environmental conditions or strategic priorities may lead organisations to adopt alternative responses.

Future research could build upon this work by comparing resilience strategies across multiple industrial technology companies and by examining how digital transformation influences the relationship between competitive advantage, operating models, and long-term strategic resilience.

Conclusion

This paper examined how an industrial technology company can strengthen its strategic resilience in response to major external disruptions. Using the case of Company A, the analysis showed that the COVID-19 pandemic did not fundamentally threaten the company's competitive position or long-term market prospects. Instead, it exposed structural dependencies within its operating model, particularly its reliance on international mobility and physical service delivery.

A structured assessment of the external environment, industry dynamics, internal capabilities, and strategic fit identified three complementary priorities for strengthening resilience: diversifying into adjacent markets, increasing customer autonomy, and expanding data-driven services. Rather than recommending a fundamental strategic transformation, the analysis supports an evolutionary approach that builds upon the company's existing strengths while reducing its exposure to future disruptions.

Beyond the specific case of Company A, this study highlights a broader lesson for industrial technology companies. Resilience depends not only on the quality of products, financial performance, or competitive positioning, but also on the adaptability of the business model through which value is created and delivered. Organizations that regularly reassess these underlying assumptions are likely to be better prepared for future uncertainty, regardless of the source of disruption.

Future research could extend this work by examining how strategic resilience evolves over time across multiple industrial organizations and by assessing the long-term impact of digital technologies on business model adaptation. Comparative studies across industries would also help determine which resilience strategies are broadly applicable and which remain context-specific.

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